

Name : S Hariram

course : CSA 0704

reg no : 1965260018

Annex: 3. Constraint-Based Problem Solving

1. DNS Resolution – Google

Open the website:

Google: <https://www.google.com/?zx=1787560655258>

Start Wireshark before opening the browser.

Constraint

You are allowed to inspect **only DNS packets**.

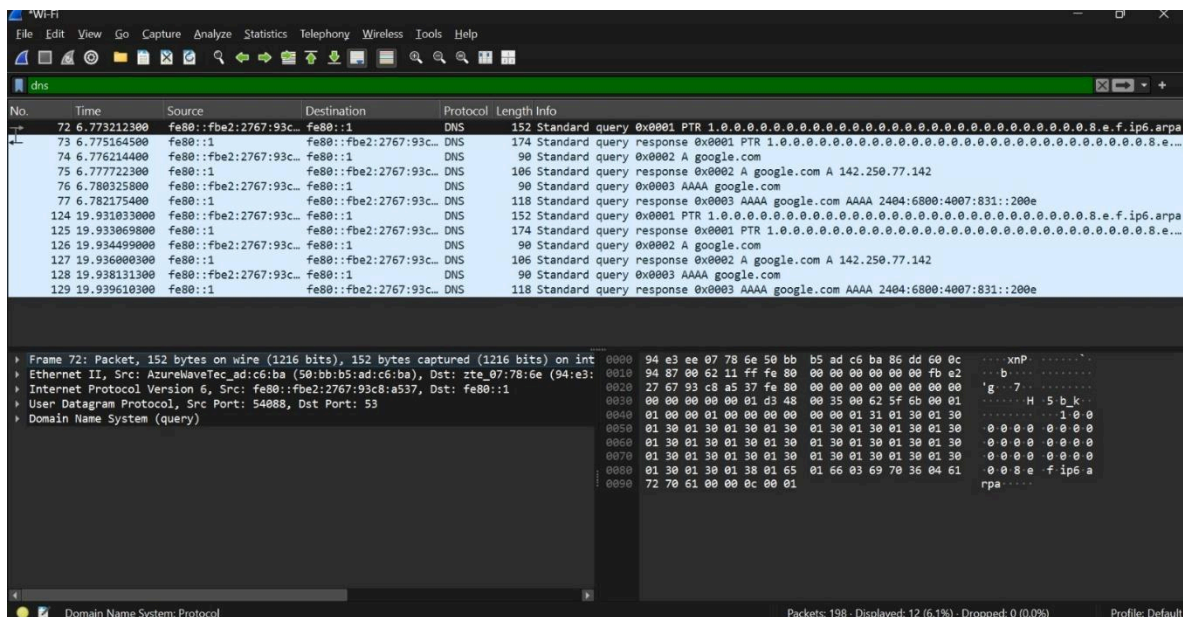
Task

Identify the packet containing the DNS response that resolves `www.google.com`.

Digital response

Select the correct packet number from the options.

Wireshark filter: Packet 75



2. HTTPS Web Access – Wikipedia

Clear the browser cache, start Wireshark, and open:

Wikipedia: <https://www.wikipedia.org/>

Constraint

You may use only **one display filter** and cannot inspect browser developer tools.

Task

Identify the transport-layer protocol carrying the encrypted web connection.

Options

- TCP
- UDP
- ICMP
- ARP

Correct answer: TCP

3. FTP Traffic Challenge – Real FTP Test Service

Use a public FTP test service provided for testing:

Rebex FTP Test Server: <https://test.rebex.net/>

Capture the connection using an FTP client.

Constraint

You may use only one port-based filter.

Task

Identify the FTP control connection.

Options

- Port 21
- Port 25
- Port 53
- Port 161

Wireshark filter: Port 21

4. SMTP Traffic Identification – Mailtrap Sandbox

Use an SMTP testing environment such as:

Mailtrap Email Sandbox: <https://mailtrap.io/email-sandbox/>

Configure the provided sandbox SMTP credentials in a mail-testing client or application and capture the connection.

Constraint

Students can inspect only packets belonging to the configured SMTP connection.

Task

Identify the application protocol responsible for transmitting the outgoing email.

Options

- SMTP
- POP  DNS
- SNMP

Correct response: SMTP